Algebra
Home Work 5

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A) Let G be a group and $H < G$ a subgroup. We assume that $G/H = H\backslash G$, i.e., every left H-coset is also a right H-coset and vice versa.

(1) Show that for every $g \in G$, there exists $g' \in G$ such that $g \cdot H = H \cdot g'$.

(2) Let g and g' be as in (1). Show that $1 \in H \cdot (g'g^{-1})$.

(3) Use (2) to deduce that H is a normal subgroup of G.

(4) As an application, show the following: if $H < G$ is a subgroup such that $\text{card}(G/H) = 2$, then H is normal in G.

Answer:

(1) Every element of G/H has the form $g.H$ for some $g \in G$. Also every element in $H\backslash G$ has the form $H.k$ for some $k \in G$. We are given that the two collections of cosets are the same. This means given $g.H$ in the set of left-cosets there will be a corresponding right-coset (and vice versa) - which we can write as $H.g'$ for some $g' \in G$.

(2) Using the notation from (1):

$$g.H = H.g' \implies g.H.g^{-1} = H.(g'g^{-1})$$

As $H < G \implies 1 \in H$. Consequently:

$$g1g^{-1} = 1 \in H.(g'g^{-1}) \tag{A.1}$$

(3) From (A.1) we have: $1 \in H.(g'g^{-1}) \implies g \in H.g'$. However, any coset is defined by any of its representative elements and hence: $H.g = H.g'$. Taking into account with $g.H = H.g'$ we must conclude that:

$$g.H = H.g \qquad (\text{true } \forall g \in G)$$

As this must hold in all cases (i.e. for all left and right cosets) we must have that $H \triangleleft G$.

(4) If $\text{Card}(G/H) = 2$. We know that the number of left (and right cosets) is 2.
H itself will be BOTH a left AND a right coset. The remaining elements from $G \setminus H$ (set difference) must then reside in the remaining left coset and also in the remaining right coset.

So we have two left cosets: $(H, g.H)$ - for some $g \in G$, and two righ cosets: $(H, H.g')$ - for some $g' \in G$.

We, therefore, must have that: $g.H = H.g'$. The conclusion is that $H \triangleleft G$.

B) Let S_3 be the symmetric group on 3 elements, i.e., the group of bijections from $\{1, 2, 3\}$ to itself.

(1) What is the cardinality of S_3?

(2) Show that the center of S_3 is trivial, i.e., that $Z(S_3) = \{\mathrm{id}_{\{1,2,3\}}\}$.

(3) Let $\sigma \in S_3$ be the element given by $\sigma(1) = 2$, $\sigma(2) = 3$ and $\sigma(3) = 1$. What is the order of σ?

(4) Using Exercise A), show that $\langle\sigma\rangle$ is normal in S_3.

(5) Show that $S_3/\langle\sigma\rangle$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

(6) Using the previous questions, compute the derived series of S_3 and show that S_3 is solvable.

Answer:

(1) $\mathrm{Card}(S_3) = 3! = 6$ - this is the total number of permutations of 3 elements.

(2) The centre of $S_3 = \{s \in S_3 \mid gs = sg;\ \forall g \in S_3\}$.

Consider the element $(12) \in S_3$ - if $s \in Z(S_3)$ then:

$$s(1\ 2) = (1\ 2)s$$

As 3 is "not moved" that means that $s(1) \neq 3$ and $s(2) \neq 3$ and so $s(3) = 3$.
Application of the same logic shows that: $s(1) = 1; s(2) = 2$ and hence $s = id$.

(3) We are given that:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

Clearly: $\sigma^3 = id$ and hence: $\mathrm{ord}(\sigma) = 3$.

(4) $\langle\sigma\rangle = \{id, \sigma, \sigma^2\} < S_3$.

Now $\langle\sigma\rangle$ has order 3 and S_3 has order 6. From this we can conclude that $[S_3 : \langle\sigma\rangle] = 2$.

Part (4) of question (A) means that $\langle\sigma\rangle \triangleleft S_3$.

(5) $S_3/\langle\sigma\rangle < S_3$ is a group with 2 elements. All groups of order 2 are all isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

(6)

C) The goal of this exercise is to show that the notions of left and right actions are equivalent. We fix a group G and a set X.

(1) Let $a : G \times X \to X$ be a left action of G on X. Show that the map $b : X \times G \to X$, given by $b(x, g) = a(g^{-1}, x)$ for $x \in X$ and $g \in G$, defines a right action of G on X.

(2) Conversely, let $b : X \times G \to X$ be a right action. Show that the map $a : G \times X \to X$, given by $a(g, x) = b(x, g^{-1})$ for $x \in X$ and $g \in G$, defines a left action of G on X.

(3) Show that the constructions in (1) and (2) are inverses of each other and define bijections between the sets of right and left actions of G on X.

Answer:

(1) We have a left G-action on X - represented by "$*$" - so $g * x$ is an example where $g \in G, x \in X$. We are defining a right G-action on X - represented by "." - which is defined by:

$$x.g = g^{-1} * x \qquad \forall x \in X, \forall g \in G$$

Now we check that it does actually represent a valid (right) G-action.

$$x.1 = 1^{-1} * x = x \qquad \forall x \in X \quad \text{(since the left G-action is valid)} \tag{C.1}$$

$$\begin{aligned} x.(g_1 g_2) &= (g_1 g_2)^{-1} * x \\ &= (g_2^{-1} g_1^{-1}) * x \\ &= g_2^{-1} * (g_1^{-1} * x) \\ &= (x.g_1).g_2 \end{aligned} \tag{C.2}$$

Equations (C.1) and (C.2) prove that we have created a valid right G-action based on a valid left G-action.

(2) Following the same pattern as in (1).
In this case we have a right G-action on X - represented by ".". We define the left G-action by $g * x$ where the left G-action is defined, via thr right G-action, by:

$$g * x = x.g^{-1} \qquad \forall x \in X, \forall g \in G$$

Now we check that it does actually represent a valid (left) G-action.

$$1 * x = x.1^{-1} = x \qquad \forall x \in X \quad \text{(since the right G-action is valid)} \tag{C.3}$$

$$\begin{aligned} (g_1 g_2) * x &= x.(g_1 g_2)^{-1} \\ &= x.(g_2^{-1} g_1^{-1}) \\ &= (x.g_2^{-1}).g_1^{-1} \\ &= g_1 * (g_2 * x) \end{aligned} \tag{C.4}$$

Equations (C.3) and (C.4) prove that we have created a valid left G-action based on a valid right G-action.

(3) Fix $g \in G$, then we can define two maps: $l_g : X \to X$, $l_g(x) = g * x$ and $r_g : X \to X$, $r_g(x) = x.g$, using the notation from above.

Both these maps are elements of $\text{Bij}(X, X)$. I will demonstrate this only for l_g as the same argument applies to r_g.

- Map is surjectice: Take any $x \in X$ and apply l_g to $(g^{-1} * x) \in X$
  We get: $l_g(g^{-1} * x) = g * (g^{-1} * x) = (gg^{-1}) * x = 1 * x = x$
- Map is injective: Assume $l_g(x) = l_g(y)$. Then:
  $\implies g * x = g * y \implies g^{-1} * (g * x) = g^{-1} * (g * y) \implies (gg^{-1}) * x = (gg^{-1}) * y \implies x = y$

Hence l_g and r_g both elements of $\text{Bij}(X, X)$.

Remembering that action "." is defined in terms of the action "$*$".
Now, for any $x \in X$, we consider $l_g[r_g(x)] = g * (x.g) = g * (g^{-1} * x) = x$. Also: $r_g[l_g(x)] = (g * x).g = g^{-1} * (g * x) = x$. Thus: $l_g \circ r_g = r_g \circ l_g = id_X$.

Hence l_g and r_g are inverse bijections of $X \to X$.

Now each $g \in G$ can be used to create two such (inverse) bijections. The above fact that l_g and r_g are inverses of each other shows that there is a direct bijection between the set of left and right actions on X.